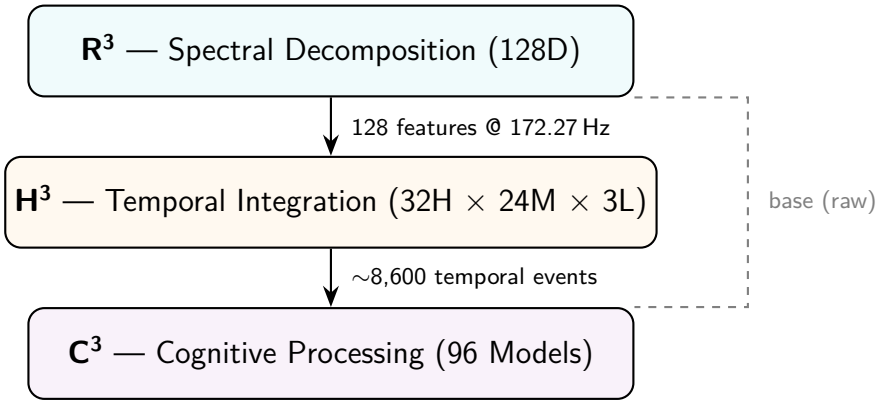
# Universal Spectral Language

## Terminology Reference

$L^3$ — Language Layer

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Musical Intelligence System

One system. One language. Spectral.

This document defines the complete terminological vocabulary used across all layers of the MI system. Every name, every address, every reference follows the conventions defined herein.

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**Core Principle**

$R^3$ names describe **what is measured** (spectral property).
$H^3$ names describe **how it evolves** (temporal morphology).
$C^3$ models **interpret** spectral evidence (emergent cognition).

No music-theory term appears as a computational primitive in $R^3$ or $H^3$.
"Chord", "key", "cadence" are $C^3$-emergent concepts.

## $R^3$ — Spectral Vocabulary (128 Features)

The complete $R^3$ spectral representation comprises 128 normalised features organised into 11 groups (A–K). Each feature is computed per analysis frame and mapped to $[0, 1]$. The *Legacy Name* column shows the original internal code identifier; the *USL Name* column shows the standardised Universal Spectral Language identifier used throughout this document.

Table 1: Complete $R^3$ feature index (128 dimensions).

| Idx | Group | Legacy Name | USL Name | Spectral Definition | Range |
|---|---|---|---|---|---|
| **Group A: Spectral Consonance [0:6]** | | | | | |
| 0 | A | roughness | roughness | Plomp–Levelt critical band interaction | $[0, 1]$ |
| 1 | A | sethares_dissonanc | sethares_dissonance | Partial-pair dissonance sum (Sethares 1993) | $[0, 1]$ |
| 2 | A | helmholtz_kang | helmholtz_fusion | Harmonic series alignment index | $[0, 1]$ |
| 3 | A | stumpf_fusion | stumpf_fusion | Perceptual fusion of simultaneous partials | $[0, 1]$ |
| 4 | A | sensory_pleasantne | sensory_pleasantnes | Composite spectral smoothness measure | $[0, 1]$ |
| 5 | A | inharmonicity | inharmonicity | Deviation of partials from harmonic series | $[0, 1]$ |
| 6 | A | harmonic_deviation | partial_deviation | Weighted partial frequency deviation | $[0, 1]$ |
| **Group B: Spectral Energy [7:11]** | | | | | |
| 7 | B | amplitude | amplitude | RMS spectral energy | $[0, 1]$ |
| 8 | B | velocity_A | amplitude_velocity | $d(\text{amplitude})/dt$ first difference | $[0, 1]$ |
| 9 | B | acceleration_A | amplitude_accelerat | $d^2(\text{amplitude})/dt^2$ second difference | $[0, 1]$ |
| 10 | B | loudness | loudness | Perceptual loudness (Stevens) | $[0, 1]$ |
| 11 | B | onset_strength | onset_strength | Spectral flux thresholded for onset detection | $[0, 1]$ |
| **Group C: Spectral Timbre [12:20]** | | | | | |
| 12 | C | warmth | spectral_warmth | Low-frequency spectral energy ratio | $[0, 1]$ |
| 13 | C | sharpness | spectral_sharpness | High-frequency weighted centroid (Zwicker) | $[0, 1]$ |
| 14 | C | tonalness | spectral_periodicit | Degree of harmonic structure in spectrum | $[0, 1]$ |
| 15 | C | clarity | spectral_clarity | Peak-to-average spectral energy ratio | $[0, 1]$ |
| 16 | C | spectral_smoothnes | spectral_smoothness | Inverse spectral irregularity | $[0, 1]$ |
| 17 | C | spectral_autocorre | spectral_autocorrel | Spectral self-similarity | $[0, 1]$ |
| 18 | C | tristimulus1 | tristimulus_low | Fundamental partial energy ratio | $[0, 1]$ |
| 19 | C | tristimulus2 | tristimulus_mid | Partials 2–4 energy ratio | $[0, 1]$ |
| 20 | C | tristimulus3 | tristimulus_high | Partials 5+ energy ratio | $[0, 1]$ |
| **Group D: Spectral Change [21:24]** | | | | | |
| 21 | D | spectral_flux | spectral_flux | Frame-to-frame $L^2$ spectral distance | $[0, 1]$ |
| 22 | D | distribution_entro | spectral_entropy | Shannon entropy of spectral distribution | $[0, 1]$ |

*Continued on next page*

*Table 1 continued from previous page*

| Idx | Group | Legacy Name | USL Name | Spectral Definition | Range |
|---|---|---|---|---|---|
| 23 | D | distribution_flatn | spectral_flatness | Geometric/arithmetic mean ratio of spectrum | [0, 1] |
| 24 | D | distribution_conce | spectral_concentrat | Spectral energy concentration measure | [0, 1] |
| **Group E: Spectral Interactions [25:48]** | | | | | |
| 25–32 | E | x_l0l5_0..7 | energy_consonance_0 | Energy × Consonance cross-product (8 dims) | [0, 1] |
| 33–40 | E | x_l4l5_0..7 | change_consonance_0 | Change × Consonance cross-product (8 dims) | [0, 1] |
| 41–48 | E | x_l5l7_0..7 | consonance_timbre_0 | Consonance × Timbre cross-product (8 dims) | [0, 1] |
| **Group F: Pitch-Class Distribution [49:64]** | | | | | |
| 49 | F | chroma_C | pce_0 | Spectral energy in pitch class 0 (C) | [0, 1] |
| 50 | F | chroma_Db | pce_1 | Spectral energy in pitch class 1 (C♯/D♭) | [0, 1] |
| 51 | F | chroma_D | pce_2 | Spectral energy in pitch class 2 (D) | [0, 1] |
| 52 | F | chroma_Eb | pce_3 | Spectral energy in pitch class 3 (D♯/E♭) | [0, 1] |
| 53 | F | chroma_E | pce_4 | Spectral energy in pitch class 4 (E) | [0, 1] |
| 54 | F | chroma_F | pce_5 | Spectral energy in pitch class 5 (F) | [0, 1] |
| 55 | F | chroma_Gb | pce_6 | Spectral energy in pitch class 6 (F♯/G♭) | [0, 1] |
| 56 | F | chroma_G | pce_7 | Spectral energy in pitch class 7 (G) | [0, 1] |
| 57 | F | chroma_Ab | pce_8 | Spectral energy in pitch class 8 (G♯/A♭) | [0, 1] |
| 58 | F | chroma_A | pce_9 | Spectral energy in pitch class 9 (A) | [0, 1] |
| 59 | F | chroma_Bb | pce_10 | Spectral energy in pitch class 10 (A♯/B♭) | [0, 1] |
| 60 | F | chroma_B | pce_11 | Spectral energy in pitch class 11 (B) | [0, 1] |
| 61 | F | pitch_height | spectral_centroid_l | $\log_2$-weighted spectral centroid | [0, 1] |
| 62 | F | pitch_class_entrop | pce_entropy | Shannon entropy of 12D PCE distribution | [0, 1] |
| 63 | F | pitch_salience | pce_salience | Peak-to-median ratio in PCE distribution | [0, 1] |
| 64 | F | inharmonicity_inde | spectral_inharmonic | 1 − (PCE peak/PCE sum) | [0, 1] |
| **Group G: Onset Periodicity [65:74]** | | | | | |
| 65 | G | tempo_estimate | onset_period | Dominant period in onset autocorrelation | [0, 1] |
| 66 | G | beat_strength | onset_autocorr_peak | Autocorrelation value at dominant period | [0, 1] |
| 67 | G | pulse_clarity | onset_periodicity_c | Dominant peak / median ratio in onset autocorr | [0, 1] |
| 68 | G | syncopation_index | onset_offbeat_ratio | Onset energy at off-period positions | [0, 1] |
| 69 | G | metricality_index | onset_subdivision_d | Count of significant nested subdivisions | [0, 1] |
| 70 | G | isochrony_nPVI | onset_interval_regu | 1 − nPVI of inter-onset intervals | [0, 1] |
| 71 | G | groove_index | onset_bass_pulse_pr | offbeat × bass_energy × periodicity_clarity | [0, 1] |
| 72 | G | event_density | onset_density | Onset count per second (normalised) | [0, 1] |
| 73 | G | tempo_stability | onset_period_stabil | 1 − CV of local onset periods | [0, 1] |
| 74 | G | rhythmic_regularit | onset_interval_entr | 1− normalised IOI histogram entropy | [0, 1] |
| **Group H: Spectral Harmony [75:86]** | | | | | |
| 75 | H | key_clarity | pcp_correlation_peal | Max correlation with 24 K–K profiles | [0, 1] |
| 76 | H | tonnetz_fifth_x | pcg_fifth_x | Pitch-class geometry: fifths circle cos | [0, 1] |
| 77 | H | tonnetz_fifth_y | pcg_fifth_y | Pitch-class geometry: fifths circle sin | [0, 1] |
| 78 | H | tonnetz_minor_x | pcg_minor_x | Pitch-class geometry: minor-3rd circle cos | [0, 1] |

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*Table 1 continued from previous page*

| Idx | Group | Legacy Name | USL Name | Spectral Definition | Range |
|---|---|---|---|---|---|
| 79 | H | tonnetz_minor_y | pcg_minor_y | Pitch-class geometry: minor-3rd circle sin | $[0, 1]$ |
| 80 | H | tonnetz_major_x | pcg_major_x | Pitch-class geometry: major-3rd circle cos | $[0, 1]$ |
| 81 | H | tonnetz_major_y | pcg_major_y | Pitch-class geometry: major-3rd circle sin | $[0, 1]$ |
| 82 | H | voice_leading_dist | chroma_l1_distance | $L^1$ norm of successive PCE differences / 2 | $[0, 1]$ |
| 83 | H | harmonic_change | chroma_frame_distan | 1− cosine similarity of successive PCE | $[0, 1]$ |
| 84 | H | tonal_stability | chroma_temporal_con | pcp_corr_peak × (1− smoothed chroma_dist) | $[0, 1]$ |
| 85 | H | diatonicity | pcp_template_fit | Max fit to diatonic pitch-class template | $[0, 1]$ |
| 86 | H | syntactic_irregula | chroma_reference_di | KL divergence from best-fit reference profile | $[0, 1]$ |
| **Group I: Spectral Information [87:93]** | | | | | |
| 87 | I | melodic_entropy | pce_transition_entr | Entropy of pitch-class transition distribution | $[0, 1]$ |
| 88 | I | harmonic_entropy | chroma_distribution_ | $\mathrm{KL}(\mathrm{PCE}_t \,\|\, \mathrm{PCE}_{\mathrm{avg}})$ exponential mapping | $[0, 1]$ |
| 89 | I | rhythmic_informati | onset_interval_surp | $-\log(p(\mathrm{IOI}))$ from running IOI histogram | $[0, 1]$ |
| 90 | I | spectral_surprise | spectral_distributi | $\mathrm{KL}(\mathrm{mel}_t \,\|\, \mathrm{mel}_{\mathrm{avg}})$ exponential mapping | $[0, 1]$ |
| 91 | I | information_rate | spectral_mutual_inf | $\mathrm{MI}(\mathrm{mel}_t; \mathrm{mel}_{t-1})/\log(128)$ | $[0, 1]$ |
| 92 | I | predictive_entropy | spectral_prediction_ | $0.5 \cdot \log(2\pi e \cdot \sigma^2_{\mathrm{res}})/\log(128)$ | $[0, 1]$ |
| 93 | I | tonal_ambiguity | pce_keyed_entropy | Entropy of softmax(K–K corr × 5) | $[0, 1]$ |
| **Group J: Spectral Timbre Extended [94:113]** | | | | | |
| 94–106 | J | mfcc_1 .. 13 | mfcc_1 .. mfcc_13 | DCT-II coefficients of log-mel spectrum (13 dims) | $[0, 1]$ |
| 107–113 | J | spectral_contrast_ | spectral_contrast_1 | Peak–valley contrast in 7 octave sub-bands | $[0, 1]$ |
| **Group K: Spectral Modulation [114:127]** | | | | | |
| 114 | K | modulation_0_5Hz | mod_energy_0.5hz | Temporal modulation energy at 0.5 Hz | $[0, 1]$ |
| 115 | K | modulation_1Hz | mod_energy_1hz | Temporal modulation energy at 1 Hz | $[0, 1]$ |
| 116 | K | modulation_2Hz | mod_energy_2hz | Temporal modulation energy at 2 Hz | $[0, 1]$ |
| 117 | K | modulation_4Hz | mod_energy_4hz | Temporal modulation energy at 4 Hz | $[0, 1]$ |
| 118 | K | modulation_8Hz | mod_energy_8hz | Temporal modulation energy at 8 Hz | $[0, 1]$ |
| 119 | K | modulation_16Hz | mod_energy_16hz | Temporal modulation energy at 16 Hz | $[0, 1]$ |
| 120 | K | modulation_centroi | mod_centroid | Centroid of modulation spectrum | $[0, 1]$ |
| 121 | K | modulation_bandwid | mod_bandwidth | Bandwidth of modulation spectrum | $[0, 1]$ |
| 122 | K | sharpness_zwicker | sharpness_zwicker | Zwicker sharpness (Bark-weighted) | $[0, 1]$ |
| 123 | K | fluctuation_streng | fluctuation_strengt | Fluctuation strength (4 Hz peak) | $[0, 1]$ |
| 124 | K | loudness_a_weighte | loudness_a_weighted | A-weighted loudness | $[0, 1]$ |
| 125 | K | alpha_ratio | alpha_ratio | Energy ratio below/above 1 kHz | $[0, 1]$ |
| 126 | K | hammarberg_index | hammarberg_index | Energy peak ratio 0–2 kHz / 2–5 kHz | $[0, 1]$ |
| 127 | K | spectral_slope_0_5 | spectral_slope_0_50 | Linear regression slope of spectrum 0–500 Hz | $[0, 1]$ |

# $H^3$ Temporal Demand Space

## $R^3 \to H^3$ Dimensional Expansion

$R^3$ produces 128 features per analysis frame at 172.27 Hz. Each feature $r_i[t]$ is a *time series*: a sequence of scalar values across $T$ frames. $H^3$ takes these time series and applies **temporal morphology**—computing how each spectral property *behaves over time* at multiple temporal scales.

**The four-dimensional demand.** Every $H^3$ output is defined by a 4-tuple:

| Component | What it selects | Size |
|---|---|---|
| `r3_idx` | Which $R^3$ feature to observe | 128 |
| `horizon` | How wide a temporal window to use | 32 |
| `morph` | Which statistical descriptor to compute | 24 |
| `law` | Which temporal perspective (causal direction) | 3 |

**Theoretical vs. active space.** The full Cartesian product yields $128 \times 32 \times 24 \times 3 =$ **294,912** possible temporal demands. In practice, each $C^3$ model declares only the demands it needs, yielding **∼8,600 active demands** across all 96 models.

**Computation.** For a given demand $(i, h, m, \ell)$ at frame $t$:

1. **Select** the $R^3$ feature series: $r_i[t]$.
2. **Window** using horizon $h$: extract $n_h$ frames according to law $\ell$:
   - $\ell = \text{L0}$ (memory): $[t - n_h + 1,\ t]$ — past → present
   - $\ell = \text{L1}$ (prediction): $[t,\ t + n_h - 1]$ — present → future
   - $\ell = \text{L2}$ (integration): $[t - n_h/2,\ t + n_h/2]$ — symmetric
3. **Weight** the window with exponential kernel: $A(\Delta t) = \exp(-3|\Delta t|/H)$.
4. **Apply morph** $m$ to the weighted window → single scalar output.

**Output shape.** Each active demand produces one scalar per frame: the "temporal morphology" of that spectral feature at that time scale. The complete $H^3$ output for a given frame is the vector of all active demand values.

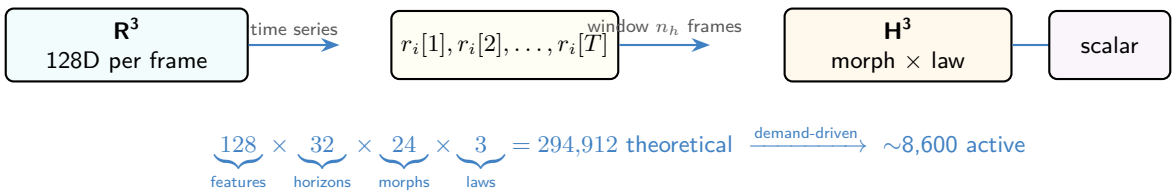


Figure 1: $R^3 \to H^3$ dimensional expansion. Each $R^3$ feature becomes a time series; $H^3$ applies temporal morphology (horizon × morph × law) to produce scalar temporal descriptors. $C^3$ models select only the demands they need from the 294,912 theoretical space.

**Concrete example.** Consider $C^3$ model EDNR (Expectancy-Driven Neural Response, PCU) requesting the demand `83.H12.M9.L0`:

| Component | Value | Meaning |
|---|---|---|
| `r3_idx = 83` | `chroma_frame_distance` | Cosine distance between successive chroma vectors |
| `horizon = H12` | `beat` (525 ms, 90 frames) | Standard beat-level temporal scale |
| `morph = M9` | `velocity_mean` | Mean rate of change over the window |
| `law = L0` | `memory` (causal) | Past → present: how fast *was* chroma changing |

Result: a single scalar per frame telling EDNR "the average speed at which spectral pitch-class distribution changed over the last beat."

## H³ Output Classification: Named Temporal Properties

R³ features are raw spectral measurements. After H³ processing, each of the ∼8,600 active measurements acquires a **named temporal identity** at three levels:

| Level | What it names | Count | Example |
|---|---|---|---|
| 1. Class | R³ Group × Morph Category | 55 | *Consonance Motion* |
| 2. Type | Class + specific feature + specific morph | 3,072* | *roughness velocity* |
| 3. Instance | Type + horizon + law | ∼8,600† | `roughness.velocity.beat.memory` |

*128 features × 24 morphs = 3,072 theoretical types. †Demand-driven subset of 294,912.

The 55 **classes** (Table 2) are the top-level semantic taxonomy. Each class contains many individual measurements. For example, the class "Consonance Level" spans 7 features × 8 distribution morphs × up to 32 horizons × 3 laws.

Table 2: Level-1 taxonomy: 55 named temporal property classes (11 R³ groups × 5 morph categories). Each cell is a *class name* containing many individual measurements. Horizon and law refine each class into specific Level-3 instances (e.g. "Consonance Motion" → `roughness.velocity.beat.memory`).

| **R³ Group** | **Distribution** M0–M7 | **Dynamics** M8–M13,15,18,21 | **Periodicity** M14,17,22 | **Information** M20 | **Geometry** M16,19,23 |
|---|---|---|---|---|---|
| **A** Consonance [0:6] | Consonance Level | Consonance Motion | Consonance Periodicity | Consonance Predictability | Consonance Contour |
| **B** Energy [7:11] | Energy Level | Energy Momentum | Energy Pulse | Energy Predictability | Energy Contour |
| **C** Timbre [12:20] | Timbre Profile | Timbre Drift | Timbre Periodicity | Timbre Predictability | Timbre Contour |
| **D** Change [21:24] | Change Level | Change Acceleration | Change Periodicity | Change Predictability | Change Contour |
| **E** Interactions [25:48] | Interaction Level | Interaction Dynamics | Interaction Periodicity | Interaction Predictability | Interaction Contour |
| **F** Pitch-Class [49:64] | Pitch-Class Profile | Pitch-Class Motion | Pitch-Class Periodicity | Pitch-Class Predictability | Pitch-Class Contour |
| **G** Onset Period [65:74] | Onset Stability | Onset Momentum | Onset Regularity | Onset Predictability | Onset Contour |
| **H** Harmony [75:86] | Harmonic Stability | Harmonic Motion | Harmonic Periodicity | Harmonic Predictability | Harmonic Contour |
| **I** Information [87:93] | Info Level | Info Dynamics | Info Periodicity | Info Meta-Predictability | Info Contour |
| **J** Timbre Ext. [94:113] | Texture Profile | Texture Drift | Texture Periodicity | Texture Predictability | Texture Contour |
| **K** Modulation [114:127] | Modulation Level | Modulation Dynamics | Modulation Periodicity | Modulation Predictability | Modulation Contour |

**Reading the matrix.** Each cell is a **named temporal property class**. Adding a horizon and law makes it fully specific. Examples:

| Property Class | Full Address | $C^3$ reads as |
|---|---|---|
| Consonance Level | `0.H12.M1.L0` | Mean roughness over the last beat (memory) |
| Consonance Motion | `0.H12.M8.L0` | How fast roughness is changing at beat scale |
| Harmonic Stability | `75.H16.M2.L2` | Std. dev. of key clarity over one measure (integrated) |
| Harmonic Periodicity | `83.H20.M14.L2` | Is chroma distance periodic at phrase scale? |
| Energy Momentum | `10.H12.M9.L0` | Mean velocity of loudness at beat scale (memory) |
| Onset Regularity | `70.H18.M14.L2` | Is onset interval regularity periodic over two measures? |
| Info Meta-Predictability | `92.H22.M20.L0` | Entropy of predictive entropy over a section |

**The naming rule.** Every $H^3$ output name follows:

**$H^3$ Temporal Property = $R^3$ Group Name + Morph Category Name + Horizon + Law**

`Consonance.Motion.beat.memory` = `0.H12.M8.L0`

$R^3$ decomposes → $H^3$ names → $C^3$ interprets.

## $H^3$ Horizon Table (32 Temporal Scales)

| Index | USL Label | Duration | Frames | Band | Spectral Scale |
|---|---|---|---|---|---|
| H0 | `micro_0` | 5.8 ms | 1 | Micro | Sub-neural integration |
| H1 | `micro_1` | 11.6 ms | 2 | Micro | Neural integration |
| H2 | `micro_2` | 17.4 ms | 3 | Micro | Onset detection boundary |
| H3 | `micro_3` | 23.2 ms | 4 | Micro | Phoneme boundary |
| H4 | `micro_4` | 34.8 ms | 6 | Micro | Pitch period scale |
| H5 | `micro_5` | 46.4 ms | 8 | Micro | Pitch integration scale |
| H6 | `sub_beat` | 200 ms | 34 | Micro | Sub-beat / note onset |
| H7 | `short_note` | 250 ms | 43 | Micro | Short note duration |
| H8 | `eighth` | 300 ms | 52 | Meso | Eighth-note scale |
| H9 | `fast_beat` | 350 ms | 60 | Meso | Fast beat (170 BPM) |
| H10 | `mod_beat` | 400 ms | 69 | Meso | Moderate beat (150 BPM) |
| H11 | `beat_low` | 450 ms | 78 | Meso | Lower beat (130 BPM) |
| H12 | `beat` | 525 ms | 90 | Meso | Standard beat (115 BPM) |
| H13 | `beat_slow` | 600 ms | 103 | Meso | Slow beat (100 BPM) |
| H14 | `dotted` | 700 ms | 121 | Meso | Dotted quarter scale |
| H15 | `two_beats` | 800 ms | 138 | Meso | Two-beat grouping |
| H16 | `measure` | 1,000 ms | 172 | Macro | Single measure scale |
| H17 | `measure_1h` | 1,500 ms | 259 | Macro | 1.5 measure scale |
| H18 | `two_meas` | 2,000 ms | 345 | Macro | Two-measure scale |
| H19 | `three_meas` | 3,000 ms | 517 | Macro | Three-measure scale |
| H20 | `phrase_s` | 5,000 ms | 861 | Macro | Short phrase (5 s) |
| H21 | `phrase_l` | 8,000 ms | 1,378 | Macro | Long phrase (8 s) |
| H22 | `section` | 15,000 ms | 2,584 | Macro | Section (15 s) |
| H23 | `section_l` | 25,000 ms | 4,307 | Macro | Extended section (25 s) |
| H24 | `movement_s` | 36 s | 6,202 | Ultra | Short movement |
| H25 | `one_min` | 60 s | 10,336 | Ultra | One minute |
| H26 | `two_min` | 120 s | 20,672 | Ultra | Two minutes |
| H27 | `movement_m` | 200 s | 34,453 | Ultra | Medium movement |
| H28 | `movement_l` | 414 s | 71,319 | Ultra | Standard movement |
| H29 | `movement_x` | 600 s | 103,359 | Ultra | Long movement |
| H30 | `form` | 800 s | 137,812 | Ultra | Extended form |
| H31 | `full_work` | 981 s | 168,999 | Ultra | Complete work |

## $H^3$ Morph Table (24 Temporal Descriptors)

| Index | USL Name | Formula | Category | Min Frames | Output Range |
|---|---|---|---|---|---|
| M0 | `attn_mean` | $\sum(A \cdot x)/\sum(A)$ | Distribution | 1 | $[0, 1]$ |
| M1 | `mean` | $\sum(x)/N$ | Distribution | 1 | $[0, 1]$ |
| M2 | `std` | $\sqrt{\mathrm{var}(x)}$ | Distribution | 2 | $[0, \infty)$ |
| M3 | `median` | middle value | Distribution | 1 | $[0, 1]$ |
| M4 | `max` | $\max(x)$ | Distribution | 1 | $[0, 1]$ |
| M5 | `range` | $\max(x) - \min(x)$ | Distribution | 2 | $[0, 1]$ |
| M6 | `skewness` | $\mu_3/\sigma^3$ | Distribution | 3 | $(-\infty, \infty)$ |
| M7 | `kurtosis` | $\mu_4/\sigma^4$ | Distribution | 4 | $[1, \infty)$ |
| M8 | `velocity` | $dx/dt$ | Dynamics | 2 | $(-\infty, \infty)$ |
| M9 | `velocity_mean` | $\mathrm{mean}(dx/dt)$ | Dynamics | 2 | $(-\infty, \infty)$ |
| M10 | `velocity_std` | $\mathrm{std}(dx/dt)$ | Dynamics | 2 | $[0, \infty)$ |
| M11 | `acceleration` | $d^2x/dt^2$ | Dynamics | 3 | $(-\infty, \infty)$ |
| M12 | `accel_mean` | $\mathrm{mean}(d^2x/dt^2)$ | Dynamics | 3 | $(-\infty, \infty)$ |
| M13 | `accel_std` | $\mathrm{std}(d^2x/dt^2)$ | Dynamics | 3 | $[0, \infty)$ |
| M14 | `periodicity` | $\max(\mathrm{autocorr}[\mathrm{lag} > 0])$ | Rhythm | 4 | $[0, 1]$ |
| M15 | `smoothness` | $1/(1 + \lvert\mathrm{jerk}\rvert/\sigma)$ | Dynamics | 3 | $(0, 1]$ |
| M16 | `curvature` | $\mathrm{mean}(\lvert d^2x/dt^2\rvert)$ | Symmetry | 3 | $[0, \infty)$ |
| M17 | `shape_period` | velocity zero-crossing $\Delta$ | Rhythm | 4 | $[0, \infty)$ |
| M18 | `trend` | OLS regression slope | Dynamics | 2 | $(-\infty, \infty)$ |
| M19 | `stability` | $1/(1 + \mathrm{var}/\sigma^2)$ | Symmetry | 2 | $(0, 1]$ |
| M20 | `entropy` | $-\Sigma\, p \cdot \log(p)\ /\ \log(16)$ | Information | 4 | $[0, 1]$ |
| M21 | `zero_cross` | sign-change count | Dynamics | 2 | $[0, N-1]$ |
| M22 | `peaks` | local maxima count | Rhythm | 3 | $[0, N-2]$ |
| M23 | `symmetry` | $\mathrm{corr}(x, \mathrm{reverse}(x))$ | Symmetry | 2 | $[-1, 1]$ |

## H³ Law Table (3 Temporal Perspectives)

| Index | USL Name | Direction | Kernel | Window |
|---|---|---|---|---|
| L0 | `memory` | Past → Present | $A(dt) = \exp(-3\lvert dt\rvert/H)$, causal | $[t-n+1,\ t]$ |
| L1 | `prediction` | Present → Future | $A(dt) = \exp(-3\lvert dt\rvert/H)$, anti-causal | $[t,\ t+n-1]$ |
| L2 | `integration` | Past ↔ Future | $A(dt) = \exp(-3\lvert dt\rvert/H)$, symmetric | $[t-n/2,\ t+n/2]$ |

# Universal Spectral Address System

Every datum flowing through the MI pipeline carries a *spectral address*: a four-component tuple that uniquely identifies its origin in the $R^3$–$H^3$ feature–temporal lattice.

```
{r3_idx}.{horizon}.{morph}.{law}
```

**Numeric example:** `83.H12.M9.L0`
**Spectral label:** `chroma_frame_distance.beat.velocity_mean.memory`

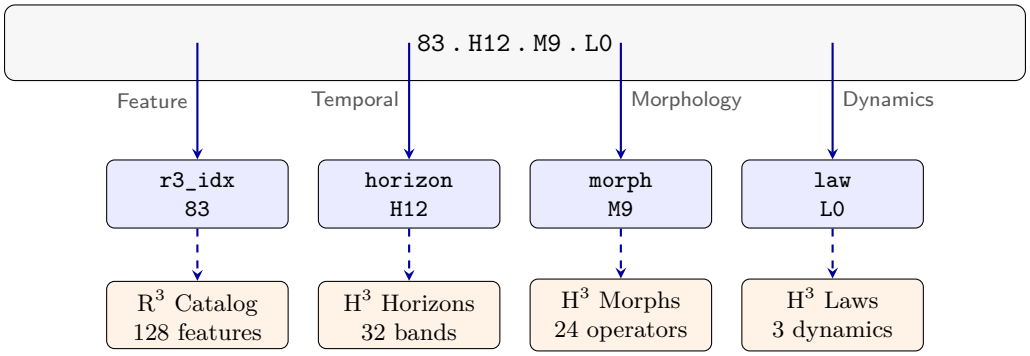


Figure 2: Four-tuple spectral address decomposition. Each component maps to a specific catalog layer in the $R^3$/$H^3$ system.

## Address Construction Rules

Table 5: Address component construction rules.

| Component | Source | Range | Example |
|---|---|---|---|
| `r3_idx` | $R^3$ Feature Index | 0–127 | 83 |
| `horizon` | $H^3$ Horizon Index | H0–H31 | H12 |
| `morph` | $H^3$ Morph Index | M0–M23 | M9 |
| `law` | $H^3$ Law Index | L0–L2 | L0 |

## Spectral Label Construction

Table 6: Spectral label construction from address components.

| Component | Format | Example |
|---|---|---|
| Feature | $R^3$ USL name | `chroma_frame_distance` |
| Horizon | $H^3$ USL label | `beat` |
| Morph | $H^3$ morph name | `velocity_mean` |
| Law | $H^3$ law name | `memory` |
| Full label | `feature.horizon.morph.law` | `chroma_frame_distance.beat.velocity_mean.memory` |

# $C^3$ Unit Interface Table

Table 7 lists all nine $C^3$ processing units and their primary interfaces with the $R^3$ feature groups and $H^3$ temporal bands. $R^3$ groups are identified by letter (A = Consonance, B = Energy, C = Timbre, D = Change, E = Interactions, F = Pitch-Class, G = Onset Period, H = Harmony, I = Information, J = Timbre Ext., K = Modulation). $H^3$ bands span Micro (<50 ms), Meso (50 ms–2 s), and Macro (>2 s) perceptual time scales.

Table 7: $C^3$ unit interface summary.

| Unit | Full Name | Models | Primary $R^3$ Groups | Primary $H^3$ Bands | Primary Laws |
|---|---|---|---|---|---|
| SPU | Sensory Processing | 9 | A, B, C, D | Micro–Meso | L0, L2 |
| STU | Structural Timing | 20 | G, H, I | Meso–Macro | L0, L1, L2 |
| IMU | Integrative Memory | 10 | F, H, I | Meso–Macro | L0, L2 |
| ASU | Auditory Scene | 9 | A, B, C | Micro–Meso | L2 |
| NDU | Neuromodulation | 8 | B, H, I | Meso–Macro | L0, L1, L2 |
| MPU | Motor Planning | 11 | G, H | Meso | L0, L1 |
| PCU | Predictive Coding | 10 | F, H, I | Micro–Macro | L0, L1, L2 |
| ARU | Arousal Regulation | 9 | B, H, I | Meso–Macro | L0, L1, L2 |
| RPU | Reward Processing | 10 | H, I, K | Meso–Macro | L0, L1, L2 |

## Provenance Chain

Figure 3 traces the complete data-flow from raw audio to the $C^3$ input package. Every tensor element retains a spectral address linking it back to its $R^3$ source through an unbroken provenance chain.

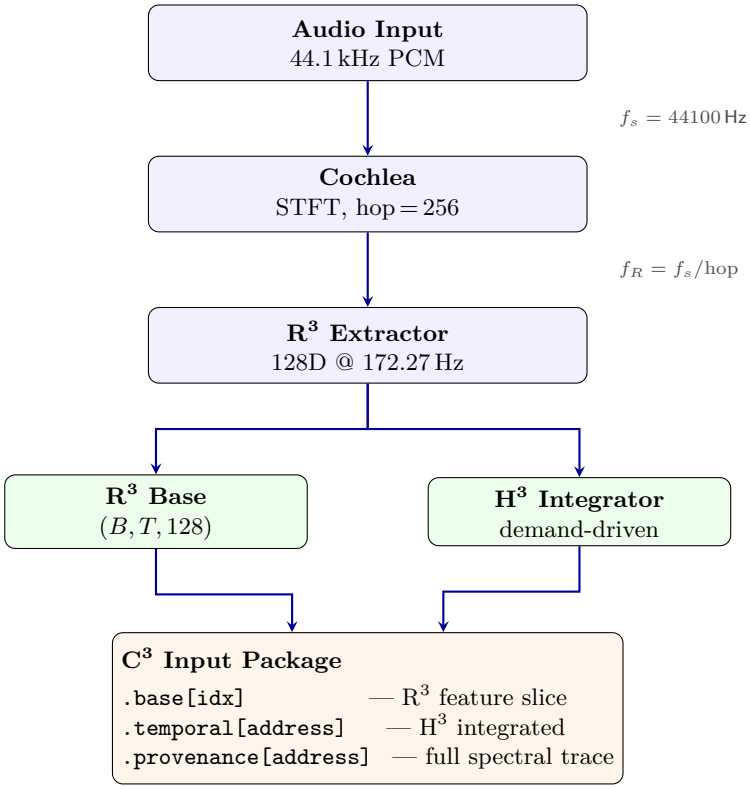


Figure 3: Provenance chain from audio input to $C^3$ input package. Every tensor element traces back to its $R^3$ source through spectral addresses.

## Naming Convention Rules

The following rules govern all naming within the Universal Spectral Language. Violations break the provenance chain and are treated as specification errors.

1. **Measurement, not interpretation.** $R^3$ feature names describe what is *measured* (e.g. `spectral_centroid`), never what is *perceived* or *interpreted*.

2. **No music-theory terms in $R^3/H^3$.** Names must not contain `chord`, `key`, `cadence`, `tonal`, or `harmonic` in their music-theoretic senses. `harmonic` is permitted only in its signal-processing meaning (integer-multiple partials).

3. **Perceptual time scales, not musical structure.** $H^3$ horizon labels denote perceptual durations. For example, `beat` designates the $\sim$525 ms periodicity band, not a musical beat or metrical position.

4. **Dot-separated addresses.** Full spectral addresses use dot separation: `feature.horizon.morph.law`. No spaces, no slashes, no camelCase.

5. **$C^3$ music-theory mapping.** $C^3$ model documentation *may* reference music-theory terms for explanatory purposes, but every such reference *must* map to one or more spectral addresses.

6. **Unbroken spectral chain.** Every $C^3$ input must trace back to an $R^3$ source through the address system. No datum enters $C^3$ without a provenance record.

# Complete $H^3$ Routing Map

This section documents every data path from $R^3$ spectral features through $H^3$ temporal morphology to $C^3$ cognitive models. The routing is **asymmetric** (not every feature uses every horizon/morph/law) but **deterministic** (each $C^3$ model declares its exact demands).

## Global Statistics

Table 8: $H^3$ demand overview (extracted from 86 model docs).

| Metric | Value |
|---|---|
| Models with explicit demands | 86 of 96 |
| Models without (ARU, older format) | 10 |
| Total tuple declarations | 1,737 |
| Unique tuples (global) | 746 |
| v1 tuples ($R^3$ [0:48]) | 532 |
| v2 tuples ($R^3$ [49:127]) | 214 |
| Theoretical space | 294,912 |
| Active fraction | 0.25% |
| $R^3$ features used (v1) | 33 of 49 |
| $R^3$ features unused | 16 (all Interaction sub-indices) |
| Horizons used | 17 of 32 |
| Morphs used | 19 of 24 |
| Laws used | 2 of 3 (L0, L2; L1 absent) |

**Notable findings.**

- **L1 (prediction) is never used.** All 1,737 demand tuples use either L0 (memory, 41%) or L2 (integration, 59%). L1 is available in the architecture but no $C^3$ model currently requests predictive-direction morphology.
- **Interaction sub-indices are sparse.** Of the 24 interaction features [25:48], only four are used: `x_l0l5[0]` (idx 25), `x_l4l5[0]` (idx 33), `x_l4l5[4]` (idx 37, ASU only), and `x_l5l7[0]` (idx 41). The remaining 20 interaction sub-indices are architecturally available but currently undemanded.
- **Top 3 features** by demand count: `spectral_flux` [10] (157 declarations), `x_l0l5[0]` [25] (144), `spectral_change` [21] (100).
- **H3 and H16 dominate** horizon usage with 433 and 442 declarations respectively, reflecting the primacy of 100 ms and 1,000 ms integration windows.

## Per-Unit Demand Summary

Table 9: Per-unit $H^3$ demand summary.

| Unit | Models | Decl. | Unique | $R^3$ feat. | Horizons | Morphs | Laws |
|---|---|---|---|---|---|---|---|
| SPU | 9 | 158 | 101 | 25 | 14 | 15 | L0, L2 |
| STU | 14 | 298 | 179 | 23 | 13 | 17 | L0, L2 |
| IMU | 15 | 347 | 209 | 23 | 15 | 17 | L0, L2 |
| ASU | 9 | 158 | 101 | 17 | 12 | 14 | L0, L2 |
| NDU | 9 | 181 | 86 | 17 | 8 | 13 | L0, L2 |
| MPU | 10 | 177 | 78 | 11 | 9 | 10 | L0, L2 |
| PCU | 10 | 215 | 111 | 16 | 10 | 14 | L0, L2 |
| ARU | 10 | *(qualitative demand; explicit tuples pending)* | | | | | |
| RPU | 10 | 203 | 115 | 15 | 10 | 14 | L0, L2 |
| **Total** | **96** | **1,737** | **746** | **33** | **17** | **19** | **2** |

## Horizon Usage Distribution

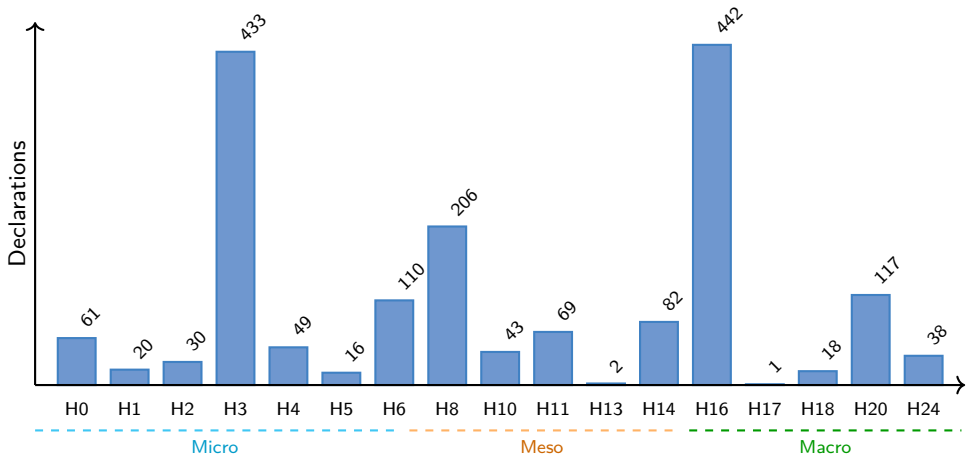


Figure 4: $H^3$ horizon usage across all 86 models. H3 (100 ms) and H16 (1,s) dominate, reflecting the two primary temporal scales of auditory cognition.

## Morph Usage Distribution

Table 10: Morph usage frequency across all 86 models. Five morphs (M7, M9, M10, M12, M23) are architecturally defined but currently undemanded by any $C^3$ model.

| Morph | Name | Count | % | Category |
|---|---|---|---|---|
| M0 | `attn_mean` | 689 | 39.7 | Distribution |
| M1 | `mean` | 428 | 24.6 | Distribution |
| M14 | `periodicity` | 146 | 8.4 | Rhythm |
| M8 | `velocity` | 93 | 5.4 | Dynamics |
| M18 | `trend` | 82 | 5.1 | Dynamics |
| M2 | `std` | 71 | 4.1 | Distribution |
| M20 | `entropy` | 58 | 3.3 | Information |
| M3 | `median` | 26 | 1.5 | Distribution |
| M4 | `max` | 27 | 1.6 | Distribution |
| M17 | `shape_period` | 25 | 1.4 | Rhythm |
| M19 | `stability` | 36 | 2.1 | Geometry |
| M15 | `smoothness` | 12 | 0.7 | Dynamics |
| M13 | `accel_std` | 12 | 0.7 | Dynamics |
| M22 | `peaks` | 11 | 0.6 | Rhythm |
| M21 | `zero_cross` | 8 | 0.5 | Dynamics |
| M5 | `range` | 6 | 0.3 | Distribution |
| M6 | `skewness` | 3 | 0.2 | Distribution |
| M11 | `acceleration` | 2 | 0.1 | Dynamics |
| M16 | `curvature` | 2 | 0.1 | Geometry |

*Unused morphs: M7 (kurtosis), M9 (velocity_mean), M10 (velocity_std), M12 (accel_mean), M23 (symmetry)*

## Law Usage

Table 11: $H^3$ law usage across all 86 models.

| Law | Name | Count | % | Status |
|---|---|---|---|---|
| L0 | Memory (past → present) | 713 | 41.0 | Active |
| L1 | Prediction (present → future) | 0 | 0.0 | **Unused** |
| L2 | Integration (past ↔ future) | 1,024 | 59.0 | Active |

**Why is L1 absent?** L1 (prediction) requires future frames, introducing latency equal to the horizon duration. All 86 models with explicit demands chose either causal (L0) or bidirectional (L2) windowing. L1 remains architecturally available for models that explicitly require predictive-only temporal perspective.

## $R^3$ Feature Routing Table (v1, [0:48])

Table 12 shows the complete routing for each v1 $R^3$ feature: which horizons, morphs, laws, and $C^3$ units consume it.

Table 12: Complete $R^3 \to H^3$ routing for all v1 features. "#H" = number of distinct horizons used; "Units" = $C^3$ units that demand this feature through $H^3$.

| Idx | Feature | #H | Horizons | #M | Morphs | Laws | Units |
|---|---|---|---|---|---|---|---|
| 0 | roughness | 10 | H0,3,6,8,10,14, 16,18,20,24 | 8 | M0,1,2,6,8, 14,18,20 | L0,L2 | SPU,STU,IMU, ASU,NDU,PCU, RPU |

*continued on next page*

| Idx | Feature | #H | Horizons | #M | Morphs | Laws | Units |
|---|---|---|---|---|---|---|---|
| 1 | sethares_diss. | 7 | H0,3,8,10,14, 16,24 | 4 | M0,1,8,19 | L0,L2 | ASU,IMU,SPU |
| 2 | helmholtz_kang | 5 | H0,3,6,8,18 | 3 | M0,1,19 | L0,L2 | IMU,SPU,STU |
| 3 | stumpf_fusion | 9 | H0,3,6,10,14, 16,18,20,24 | 7 | M0,1,3,6,8,14,19 | L0,L2 | ASU,IMU,MPU, SPU |
| 4 | sensory_pleas. | 7 | H3,8,10,16,18, 20,24 | 8 | M0,1,2,8,15, 18,19,20 | L0,L2 | ASU,IMU,NDU, PCU,RPU,SPU |
| 5 | inharmonicity | 12 | H0,1,3,5,6,8, 10,11,14,16,20,24 | 8 | M0,1,2,8,14, 18,19,22 | L0,L2 | ASU,IMU,PCU, SPU |
| 6 | harmonic_dev. | 6 | H0,3,10,14,16,20 | 3 | M0,1,4 | L0,L2 | IMU,PCU,SPU |
| 7 | amplitude | 10 | H0,1,3,6,8,11, 14,16,20,24 | 10 | M0,1,2,3,4,5, 8,15,18,20 | L0,L2 | ASU,IMU,MPU, NDU,PCU,RPU, STU |
| 8 | loudness | 7 | H3,6,8,11,14, 16,20 | 12 | M0,1,2,4,5,8, 14,15,17,18, 19,20 | L0,L2 | *all 8 units* |
| 9 | acceleration_A | 6 | H3,4,8,11,14,16 | 5 | M0,1,8,14,15 | L0,L2 | ASU,PCU,RPU, STU |
| 10 | spectral_flux | 11 | H0,1,3,4,6,8, 10,11,16,20,24 | 9 | M0,1,2,3,4,8, 14,17,19 | L0,L2 | *all 8 units* |
| 11 | onset_strength | 10 | H0,3,6,8,10,11, 13,14,16,20 | 10 | M0,1,2,4,5,8, 11,14,15,17 | L0,L2 | ASU,IMU,MPU, NDU,PCU,SPU, STU |
| 12 | warmth | 8 | H2,3,5,8,14, 16,20,24 | 4 | M0,1,3,19 | L0,L2 | IMU,MPU,RPU, SPU,STU |
| 13 | sharpness | 4 | H2,3,4,16 | 6 | M0,1,2,8,17,20 | L0,L2 | NDU,RPU,SPU, STU |
| 14 | tonalness | 13 | H0,1,2,3,4,5,6, 8,10,14,16,20,24 | 8 | M0,1,2,3,14, 18,19,22 | L0,L2 | ASU,IMU,NDU, PCU,RPU,SPU, STU |
| 15 | clarity | 5 | H3,5,8,10,18 | 4 | M0,1,2,3 | L0,L2 | ASU,IMU,MPU, SPU,STU |
| 16 | spec._smooth. | 5 | H0,3,14,16,20 | 4 | M0,1,2,20 | L0,L2 | ASU,IMU,NDU, SPU,STU |
| 17 | spec._autocorr. | 4 | H3,6,8,10 | 3 | M0,1,14 | L0,L2 | IMU,NDU,SPU, STU |
| 18 | tristimulus1 | 7 | H0,2,3,6,16,20,24 | 3 | M0,1,22 | L0,L2 | IMU,SPU,STU |
| 19 | tristimulus2 | 3 | H0,2,3 | 1 | M0 | L2 | SPU,STU |
| 20 | tristimulus3 | 3 | H0,2,3 | 1 | M0 | L2 | SPU,STU |
| 21 | spectral_change | 11 | H1,2,3,4,8,10, 11,14,16,18,20 | 14 | M0,1,2,3,4,5, 8,13,14,15,17, 18,19,20 | L0,L2 | *all 8 units* |
| 22 | energy_change | 11 | H3,4,6,8,10,11, 14,16,18,20,24 | 12 | M0,1,2,3,8,11, 13,14,15,18, 19,20 | L0,L2 | *all 8 units* |
| 23 | pitch_change | 9 | H3,4,6,8,11, 14,16,20,24 | 9 | M0,1,2,3,8, 14,18,19,20 | L0,L2 | IMU,NDU,PCU, SPU,STU |
| 24 | concentration | 5 | H3,4,8,14,16 | 6 | M0,1,2,3,19,20 | L0,L2 | NDU,RPU,SPU, STU |
| 25 | x_l0l5[0] | 11 | H0,1,3,4,6,8, 11,14,16,17,20 | 14 | M0,1,2,8,13, 14,15,16,17,18, 19,20,21,22 | L0,L2 | *all 8 units* |
| 33 | x_l4l5[0] | 7 | H3,4,8,11,14,16,20 | 10 | M0,1,2,8,14, 17,18,19,20,22 | L0,L2 | IMU,MPU,NDU, PCU,RPU,SPU, STU |
| 37 | x_l4l5[4] | 2 | H3,16 | 6 | M0,1,2,17,20,21 | L2 | ASU |
| 41 | x_l5l7[0] | 6 | H3,6,8,14,16,20 | 11 | M0,1,2,5,6,13, 14,16,18,19,20 | L0,L2 | IMU,NDU,PCU, RPU,SPU,STU |

*Indices 26–32, 34–36, 38–40, 42–48 are architecturally defined but undemanded by any model.*

## Unit–Feature Affinity Matrix

Figure 5 shows which $R^3$ feature groups are consumed by each $C^3$ unit through $H^3$. Cell shading intensity reflects the number of unique demand tuples.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| SPU | 25 | 8 | 27 | 12 | 10 |
| STU | 3 | 52 | 16 | 38 | 29 |
| IMU | 49 | 45 | 22 | 27 | 7 |
| ASU | 16 | 26 | 12 | 11 | 23 |
| NDU | 4 | 22 | 10 | 13 | 19 |
| MPU | 1 | 19 | 2 | 5 | 24 |
| PCU | 17 | 22 | 6 | 14 | 21 |
| RPU | 14 | 21 | 5 | 17 | 24 |

Cell value = number of unique tuples from that group

Figure 5: Unit–feature group affinity matrix. Cell values show the number of unique $H^3$ demand tuples from each $R^3$ group per unit. Darker shading = higher demand density.

## Per-Unit Detailed Routing

The following tables show every $R^3$ feature consumed by each unit through $H^3$, including exact horizon, morph, and law ranges.

### SPU — Sensory Processing Unit (9 models, 71 v1 tuples)

| $R^3$ | Feature | # | Horizons | Morphs | Laws |
|---|---|---|---|---|---|
| 0 | roughness | 5 | H0,3,6 | M0,1,14,18 | L0,L2 |
| 1 | sethares_diss. | 1 | H0 | M0 | L2 |
| 2 | helmholtz_kang | 4 | H0,3,6 | M0,1,19 | L0,L2 |
| 3 | stumpf_fusion | 2 | H0,6 | M0,1 | L0,L2 |
| 4 | sensory_pleas. | 1 | H3 | M0 | L2 |
| 5 | inharmonicity | 6 | H0,3,5,6 | M0,1,18 | L0,L2 |
| 6 | harmonic_dev. | 2 | H0,3 | M0,1 | L0,L2 |
| 8 | loudness | 1 | H8 | M1 | L0 |
| 11 | onset_str. | 1 | H3 | M0 | L0 |
| 12 | warmth | 2 | H2,5 | M0,1 | L0,L2 |
| 13 | sharpness | 1 | H2 | M0 | L2 |
| 14 | tonalness | 7 | H0,2,3,5,6,8 | M0,1,19 | L0,L2 |
| 15 | clarity | 2 | H3,8 | M0,1 | L0,L2 |
| 16 | spec. smooth. | 1 | H3 | M0 | L2 |
| 17 | spec. autocorr. | 3 | H3,6 | M0,14 | L2 |
| 18 | tristimulus1 | 4 | H0,2,3,6 | M0,1 | L0,L2 |
| 19 | tristimulus2 | 3 | H0,2,3 | M0 | L2 |
| 20 | tristimulus3 | 3 | H0,2,3 | M0 | L2 |
| 21 | spec. change | 3 | H3,8 | M1,8,13 | L0 |
| 22 | energy_change | 3 | H3,6,8 | M0,1,8 | L0,L2 |
| 23 | pitch_change | 3 | H3,6 | M0,1,8 | L0,L2 |
| 24 | concentration | 3 | H3,8 | M0,1,3 | L0,L2 |
| 25 | x_l0l5[0] | 2 | H3,6 | M0,14 | L2 |
| 33 | x_l4l5[0] | 2 | H8 | M0,14 | L2 |
| 41 | x_l5l7[0] | 6 | H3,6,8 | M0,1,14,19 | L0,L2 |

### STU — Structural Timing Unit (14 models, 140 v1 tuples)

| $R^3$ | Feature | # | Horizons | Morphs | Laws |
|---|---|---|---|---|---|
| 0 | roughness | 2 | H20 | M1,18 | L0 |
| 2 | helmholtz_kang | 1 | H8 | M1 | L0 |
| 7 | amplitude | 13 | H6,8,11,14,16,20 | M0,1,3,4,15,18 | L0,L2 |

| | | | | | |
|---|---|---|---|---|---|
| 8 | loudness | 15 | H6,11,14,16,20 | M0,1,4,14,15,18,19 | L0 |
| 9 | accel. A | 5 | H8,11,14,16 | M0,1,14,15 | L0,L2 |
| 10 | spec. flux | 11 | H6,8,11,16 | M0,1,4,8,14,17 | L0 |
| 11 | onset_str. | 9 | H6,8,11,16 | M0,1,4,14,15,17 | L0 |
| 12 | warmth | 3 | H2,14 | M0,1,3 | L0,L2 |
| 13 | sharpness | 1 | H2 | M0 | L2 |
| 14 | tonalness | 7 | H5,6,8,14 | M0,1,3 | L0,L2 |
| 15 | clarity | 1 | H5 | M0 | L0 |
| 16 | spec. smooth. | 1 | H14 | M1 | L0 |
| 17 | spec. autocorr. | 2 | H8 | M0,1 | L0,L2 |
| 18 | tristimulus1 | 3 | H2,20 | M0,1,22 | L0,L2 |
| 19 | tristimulus2 | 1 | H2 | M0 | L2 |
| 20 | tristimulus3 | 1 | H2 | M0 | L2 |
| 21 | spec. change | 9 | H8,11,16 | M1,3,8,14,15,17 | L0,L2 |
| 22 | energy_change | 13 | H6,8,11,14,16 | M1,3,8,11,13,14,15 | L0,L2 |
| 23 | pitch_change | 8 | H8,14,16,20 | M0,1,3,8,14,18 | L0,L2 |
| 24 | concentration | 5 | H8,14,16 | M0,1,3,19 | L0 |
| 25 | x_l0l5[0] | 12 | H11,14,16,20 | M0,1,13,14,15,19,22 | L0,L2 |
| 33 | x_l4l5[0] | 10 | H11,14,16,20 | M0,1,14,17,18,19,22 | L0,L2 |
| 41 | x_l5l7[0] | 7 | H14,16,20 | M1,13,14,18,19 | L0,L2 |

## IMU — Integrative Memory Unit (15 models, 150 v1 tuples)

| $R^3$ | Feature | # | Horizons | Morphs | Laws |
|---|---|---|---|---|---|
| 0 | roughness | 7 | H10,14,16,18,20,24 | M0,1,18 | L0,L2 |
| 1 | sethares_diss. | 4 | H10,14,16,24 | M0,1,8,19 | L0,L2 |
| 2 | helmholtz_kang | 1 | H18 | M1 | L0 |
| 3 | stumpf_fusion | 16 | H0,3,6,10,14,16,18,20,24 | M0,1,3,14,19 | L0,L2 |
| 4 | sensory_pleas. | 7 | H10,16,18,20,24 | M0,1,18,19 | L0,L2 |
| 5 | inharmonicity | 10 | H6,10,11,14,16,20,24 | M0,1,8,14,19,22 | L0,L2 |
| 6 | harmonic_dev. | 4 | H10,14,16,20 | M0,1 | L0,L2 |
| 7 | amplitude | 13 | H6,11,16,20,24 | M0,1,3,4,5,8,18 | L0,L2 |
| 8 | loudness | 5 | H6,11,16 | M0,1,8,17 | L0,L2 |
| 10 | spec. flux | 13 | H6,10,11,16,20,24 | M0,1,3,4,14,17,19 | L0,L2 |
| 11 | onset_str. | 14 | H6,10,11,14,16,20 | M0,1,4,5,8,14,17 | L0,L2 |
| 12 | warmth | 3 | H16,20,24 | M0,1,19 | L0,L2 |
| 14 | tonalness | 11 | H0,3,6,10,14,16,20,24 | M0,1,3,18,19,22 | L0,L2 |
| 15 | clarity | 2 | H10,18 | M0,3 | L0,L2 |
| 16 | spec. smooth. | 1 | H20 | M1 | L0 |
| 17 | spec. autocorr. | 3 | H3,6,10 | M14 | L0,L2 |
| 18 | tristimulus1 | 2 | H16,24 | M0,1 | L0,L2 |
| 21 | spec. change | 9 | H10,14,16,18,20 | M0,1,3,4,5,8 | L0,L2 |
| 22 | energy_change | 13 | H6,10,11,14,16,18,20,24 | M0,1,13,14,18,19 | L0,L2 |
| 23 | pitch_change | 5 | H11,14,16,20,24 | M0,1,19 | L0,L2 |
| 25 | x_l0l5[0] | 2 | H6,16 | M0,19 | L0,L2 |
| 33 | x_l4l5[0] | 2 | H11 | M0,17 | L0,L2 |
| 41 | x_l5l7[0] | 3 | H16 | M1,14,18 | L0,L2 |

## ASU — Auditory Scene Analysis Unit (9 models, 87 v1 tuples)

| $R^3$ | Feature | # | Horizons | Morphs | Laws |
|---|---|---|---|---|---|
| 0 | roughness | 6 | H0,3,16 | M0,1,2,20 | L2 |
| 1 | sethares_diss. | 2 | H3,8 | M0,8 | L0,L2 |
| 3 | stumpf_fusion | 3 | H3,6,16 | M0,6,8 | L2 |
| 4 | sensory_pleas. | 3 | H3,16 | M0,1,8 | L2 |
| 5 | inharmonicity | 2 | H3 | M0,2 | L2 |
| 7 | amplitude | 3 | H3,16 | M0,1,2 | L2 |
| 8 | loudness | 5 | H3,16 | M0,1,2,20 | L2 |
| 9 | accel. A | 1 | H3 | M0 | L2 |
| 10 | spec. flux | 10 | H0,1,3,4,8,16 | M0,1,14,17 | L2 |
| 11 | onset_str. | 7 | H0,3,13,16 | M0,1,2,8,11,14 | L0,L2 |
| 14 | tonalness | 6 | H0,1,3,4,16 | M0,1,14 | L2 |
| 15 | clarity | 2 | H3 | M0,2 | L2 |
| 16 | spec. smooth. | 3 | H0,3,16 | M0,1,20 | L2 |
| 21 | spec. change | 6 | H1,3,4,8,16 | M0,2,8,14 | L0,L2 |
| 22 | energy_change | 5 | H3,8,16 | M0,2,8 | L0,L2 |
| 25 | x_l0l5[0] | 16 | H1,3,8,16,17,20 | M0,1,2,8,14,17,18,19,20,21 | L0,L2 |
| 37 | x_l4l5[4] | 7 | H3,16 | M0,1,2,17,20,21 | L2 |

## NDU — Neuromodulation Unit (9 models, 68 v1 tuples)

| R³ | Feature | # | Horizons | Morphs | Laws |
|---|---|---|---|---|---|
| 0 | roughness | 2 | H3 | M0,20 | L2 |
| 4 | sensory_pleas. | 2 | H3,16 | M0,1 | L2 |
| 7 | amplitude | 5 | H0,3,8 | M0,1,20 | L2 |
| 8 | loudness | 6 | H3,8,16 | M0,1,5,20 | L0,L2 |
| 10 | spec. flux | 6 | H0,1,3,16 | M0,1,2,14 | L2 |
| 11 | onset_str. | 5 | H0,3,16 | M0,1,8,14 | L0,L2 |
| 13 | sharpness | 5 | H3,4 | M0,2,8,17,20 | L0,L2 |
| 14 | tonalness | 2 | H3,16 | M0,1 | L2 |
| 16 | spec. smooth. | 2 | H3,16 | M0,2 | L2 |
| 17 | spec. autocorr. | 1 | H3 | M0 | L2 |
| 21 | spec. change | 5 | H3,4,16 | M0,2,8,18,20 | L0,L2 |
| 22 | energy_change | 2 | H3 | M0,2 | L2 |
| 23 | pitch_change | 4 | H3,4,16 | M0,1,2,20 | L2 |
| 24 | concentration | 2 | H3,16 | M0,2 | L2 |
| 25 | x_l0l5[0] | 7 | H3,8,16 | M0,1,2,14 | L0,L2 |
| 33 | x_l4l5[0] | 6 | H3,16 | M0,1,2,8,18,20 | L0,L2 |
| 41 | x_l5l7[0] | 6 | H3,16 | M0,1,2,14,16,20 | L2 |

## MPU — Motor Planning Unit (10 models, 54 v1 tuples)

| R³ | Feature | # | Horizons | Morphs | Laws |
|---|---|---|---|---|---|
| 3 | stumpf_fusion | 1 | H3 | M0 | L2 |
| 7 | amplitude | 4 | H3,8,16 | M0,1,2 | L2 |
| 8 | loudness | 4 | H3,8 | M0,1,2,20 | L0,L2 |
| 10 | spec. flux | 10 | H0,1,3,4,16 | M0,1,2,14 | L0,L2 |
| 11 | onset_str. | 4 | H3,8,16 | M0,14 | L0,L2 |
| 12 | warmth | 1 | H3 | M0 | L2 |
| 15 | clarity | 1 | H3 | M0 | L2 |
| 21 | spec. change | 4 | H4,16 | M1,2,8 | L0,L2 |
| 22 | energy_change | 1 | H8 | M8 | L0 |
| 25 | x_l0l5[0] | 14 | H0,1,3,4,8,16 | M0,1,2,14,19,21 | L0,L2 |
| 33 | x_l4l5[0] | 10 | H3,8,16 | M0,1,2,8,14,19,20 | L0,L2 |

## PCU — Predictive Coding Unit (10 models, 77 v1 tuples)

| R³ | Feature | # | Horizons | Morphs | Laws |
|---|---|---|---|---|---|
| 0 | roughness | 3 | H3,8,16 | M0,1 | L0,L2 |
| 4 | sensory_pleas. | 4 | H3,8,16 | M0,1,2,20 | L0,L2 |
| 5 | inharmonicity | 7 | H0,1,3,8,16 | M0,1,14,18 | L0,L2 |
| 6 | harmonic_dev. | 3 | H0,3 | M0,4 | L2 |
| 7 | amplitude | 5 | H0,1,3,16 | M0,1,2 | L2 |
| 8 | loudness | 2 | H3,16 | M0,8 | L2 |
| 9 | accel. A | 3 | H3,4,8 | M0,1,8 | L0,L2 |
| 10 | spec. flux | 10 | H0,1,3,8,16 | M0,1,8,14 | L0,L2 |
| 11 | onset_str. | 2 | H3,16 | M0,14 | L2 |
| 14 | tonalness | 3 | H3,8,16 | M0,1 | L0,L2 |
| 21 | spec. change | 11 | H1,3,4,8,16 | M0,1,2,4,8,19,20 | L0,L2 |
| 22 | energy_change | 2 | H3,4 | M8 | L0 |
| 23 | pitch_change | 1 | H3 | M8 | L2 |
| 25 | x_l0l5[0] | 10 | H0,1,3,8,16 | M0,1,2,14,16,21 | L0,L2 |
| 33 | x_l4l5[0] | 3 | H4,8,16 | M0,1,8 | L0,L2 |
| 41 | x_l5l7[0] | 8 | H3,8,16 | M0,1,6,18,20 | L0,L2 |

## RPU — Reward Processing Unit (10 models, 81 v1 tuples)

| R³ | Feature | # | Horizons | Morphs | Laws |
|---|---|---|---|---|---|
| 0 | roughness | 7 | H3,8,16 | M0,1,2,6,8 | L0,L2 |
| 4 | sensory_pleas. | 7 | H3,8,16 | M0,1,2,8,15,18 | L0,L2 |
| 7 | amplitude | 6 | H8,16 | M0,1,2,8 | L0,L2 |
| 8 | loudness | 7 | H3,8,16,20 | M0,1,4,8,18 | L0,L2 |
| 9 | accel. A | 3 | H3,8,16 | M0,1,8 | L2 |

| | | | | | |
|---|---|---|---|---|---|
| 10 | spec. flux | 5 | H3,4,8 | M0,2,14 | L2 |
| 12 | warmth | 2 | H8,16 | M1 | L2 |
| 13 | sharpness | 1 | H16 | M1 | L2 |
| 14 | tonalness | 2 | H16 | M1,2 | L2 |
| 21 | spec. change | 9 | H2,3,4,8,16 | M0,1,8,20 | L0,L2 |
| 22 | energy_change | 4 | H8,16 | M8,18,20 | L0,L2 |
| 24 | concentration | 4 | H3,4,8,16 | M0,2,20 | L2 |
| 25 | x_l0l5[0] | 10 | H3,8,16,20 | M0,1,2,14,18,20 | L0,L2 |
| 33 | x_l4l5[0] | 8 | H3,8,16 | M0,1,2,8,18,20 | L0,L2 |
| 41 | x_l5l7[0] | 6 | H8,16 | M0,1,5,18,20 | L0,L2 |

## Per-Model Demand Counts

Table 21: $H^3$ demand tuples per model (86 models with explicit demands). ARU models (10) are omitted—their demands are specified qualitatively.

| Model | # | Model | # | Model | # | Model | # |
|---|---|---|---|---|---|---|---|
| SPU-$\alpha$1-BCH | 26 | STU-$\alpha$1-HMCE | 24 | IMU-$\alpha$1-MEAMN | 25 | ASU-$\alpha$1-SNEM | 20 |
| SPU-$\alpha$2-PSCL | 23 | STU-$\alpha$2-AMSC | 22 | IMU-$\alpha$2-PNH | 20 | ASU-$\alpha$2-IACM | 24 |
| SPU-$\alpha$3-PCCR | 20 | STU-$\alpha$3-MDNS | 22 | IMU-$\alpha$3-MMP | 23 | ASU-$\alpha$3-CSG | 21 |
| SPU-$\beta$1-STAI | 20 | STU-$\beta$1-AMSS | 18 | IMU-$\beta$1-RASN | 36 | ASU-$\beta$1-BARM | 16 |
| SPU-$\beta$2-TSCP | 16 | STU-$\beta$2-TPIO | 18 | IMU-$\beta$2-PMIM | 26 | ASU-$\beta$2-STANM | 17 |
| SPU-$\beta$3-MIAA | 17 | STU-$\beta$3-EDTA | 21 | IMU-$\beta$3-OII | 24 | ASU-$\beta$3-AACM | 13 |
| SPU-$\gamma$1-SDNPS | 12 | STU-$\beta$4-ETAM | 28 | IMU-$\beta$4-HCMC | 29 | ASU-$\gamma$1-PWSM | 17 |
| SPU-$\gamma$2-ESME | 14 | STU-$\beta$5-HGSIC | 23 | IMU-$\beta$5-RIRI | 21 | ASU-$\gamma$2-DGTP | 11 |
| SPU-$\gamma$3-SDED | 10 | STU-$\beta$6-OMS | 23 | IMU-$\beta$6-MSPBA | 25 | ASU-$\gamma$3-SDL | 19 |
| | | STU-$\gamma$1-TMRM | 19 | IMU-$\beta$7-VRIAP | 20 | | |
| | | STU-$\gamma$2-NEWMD | 20 | IMU-$\beta$8-TPRD | 23 | | |
| | | STU-$\gamma$3-MTNE | 18 | IMU-$\beta$9-CMAPCC | 20 | | |
| | | STU-$\gamma$4-PTGMP | 18 | IMU-$\gamma$1-DMMS | 19 | | |
| | | STU-$\gamma$5-MPFS | 24 | IMU-$\gamma$2-CSSL | 18 | | |
| | | | | IMU-$\gamma$3-CDEM | 18 | | |
| NDU-$\alpha$1-MPG | 17 | MPU-$\alpha$1-PEOM | 22 | PCU-$\alpha$1-HTP | 28 | RPU-$\alpha$1-DAED | 18 |
| NDU-$\alpha$2-SDD | 26 | MPU-$\alpha$2-MSR | 28 | PCU-$\alpha$2-SPH | 21 | RPU-$\alpha$2-MORMR | 17 |
| NDU-$\alpha$3-EDNR | 20 | MPU-$\alpha$3-GSSM | 20 | PCU-$\alpha$3-ICEM | 21 | RPU-$\alpha$3-RPEM | 24 |
| NDU-$\beta$1-DSP | 19 | MPU-$\beta$1-ASAP | 15 | PCU-$\beta$1-PWUP | 17 | RPU-$\beta$1-IUCP | 22 |
| NDU-$\beta$2-CDMR | 24 | MPU-$\beta$2-DDSMI | 14 | PCU-$\beta$2-WMED | 21 | RPU-$\beta$2-MCCN | 22 |
| NDU-$\beta$3-SLEE | 24 | MPU-$\beta$3-VRMSME | 12 | PCU-$\beta$3-UDP | 24 | RPU-$\beta$3-MEAMR | 14 |
| NDU-$\gamma$1-SDDP | 16 | MPU-$\beta$4-SPMC | 19 | PCU-$\beta$4-CHPI | 29 | RPU-$\beta$4-SSRI | 28 |
| NDU-$\gamma$2-ONI | 16 | MPU-$\gamma$1-NSCP | 16 | PCU-$\gamma$1-IGFE | 22 | RPU-$\gamma$1-LDAC | 20 |
| NDU-$\gamma$3-ECT | 19 | MPU-$\gamma$2-CTBB | 13 | PCU-$\gamma$2-MAA | 14 | RPU-$\gamma$2-IOTMS | 16 |
| | | MPU-$\gamma$3-STC | 18 | PCU-$\gamma$3-PSH | 18 | RPU-$\gamma$3-SSPS | 22 |

## Universal vs. Specialized Features

Five $R^3$ features are demanded by all 8 units with explicit tuples, making them **universal routing targets**:

Table 22: $R^3$ features used by all 8 units (excluding ARU).

| Idx | Feature | Group | Total Decl. | Unique Tuples |
|---|---|---|---|---|
| 8 | loudness | B: Energy | 91 | 40 |
| 10 | spectral_flux | B: Energy | 157 | 62 |
| 21 | spectral_change | D: Change | 100 | 56 |
| 22 | energy_change | D: Change | 78 | 45 |
| 25 | x_l0l5[0] | E: Interactions | 144 | 81 |

These five features represent the core "sensing" dimensions that every cognitive subsystem needs: energy dynamics (loudness, spectral flux), spectral change (spectral/energy change), and cross-feature coupling (x_l0l5).